Genetics: CDKN2A-CDK4-TP53 Pathway

Germline mutations in the chromosome 9p21 tumor suppressor gene, *CDKN2A* (cyclin-dependent kinase inhibitor 2A), account for approximately 40% of hereditary melanoma cases (defined as ≥3 melanomas in one lineage) and confer a 76% lifetime risk of developing melanoma in the United States. This risk varies by geographic region, being higher in Australia and lower in Europe, consistent with ambient UV exposure levels. Individuals with germline *CDKN2A* mutations also have an increased risk of pancreatic cancer, with an estimated 15% lifetime risk.

Although bona fide deleterious point mutations in *CDKN2A* are relatively uncommon in primary melanoma tumors, somatic inactivation—particularly through homozygous deletions—may be underrecognized due to technical limitations in detection.

The *CDKN2A* gene encodes two distinct tumor suppressor proteins through alternative reading frames: **p16** (also known as **INK4a**) and **p14$^{\mathbf{ARF}}$**.

- **p16** functions as a cell cycle regulator by binding to and inhibiting cyclin-dependent kinases **CDK4** and **CDK6**. This inhibition prevents phosphorylation of the retinoblastoma (Rb) protein, thereby blocking progression from G1 to S phase of the cell cycle.
- Loss or inactivation of p16 leads to unchecked CDK4/6 activity, resulting in Rb hyperphosphorylation, release of the E2F transcription factor, and uncontrolled entry into S phase—promoting cellular proliferation and tumorigenesis.

CDK4 is the direct binding partner of p16. Hereditary mutations in *CDK4* are rare, with only a few families reported worldwide. However, somatic *CDK4* mutations have been identified in some melanoma cell lines. These mutations typically render CDK4 resistant to p16-mediated inhibition, producing a functional outcome equivalent to p16 loss.

The second protein product of *CDKN2A*, **p14$^{\mathbf{ARF}}$**, stabilizes the **p53** tumor suppressor by inhibiting **MDM2**, an E3 ubiquitin ligase that promotes p53 degradation. Loss of p14$^{\mathrm{ARF}}$ function leads to increased MDM2 activity, accelerated p53 degradation, and impaired DNA damage response.

Thus, inactivation of *CDKN2A* disrupts both the **Rb pathway** (via p16 loss) and the **p53 pathway** (via p14$^{\mathrm{ARF}}$ loss), effectively deregulating two major tumor suppressor networks. This dual impact likely explains the relatively low frequency of direct *TP53* mutations in melanoma, as p53 function is already compromised through upstream mechanisms.

RAS Signaling Pathway

In 2002, Davies et al. reported somatic mutations in the *BRAF* gene in 66% of melanomas. *BRAF* encodes **B-raf**, a serine/threonine kinase that plays a central role in the **Ras-Raf-MEK-ERK** (MAPK) signaling pathway, which regulates cell growth, proliferation, and differentiation in response to extracellular signals such as growth factors, cytokines, and hormones.

Activation of this pathway proceeds as follows: 1. Extracellular ligands activate receptor tyrosine kinases. 2. This triggers **Ras** (a G-protein) to exchange GDP for GTP, activating it. 3. Active Ras recruits and activates **Raf** (including B-raf). 4. Raf phosphorylates and activates **MEK**, which then phosphorylates **ERK** (extracellular signal-regulated kinase). 5. Activated ERK translocates to the nucleus and regulates transcription factors involved in cell cycle progression.

Humans express three Raf isoforms: **A-raf**, **B-raf**, and **C-raf**. Among these, **B-raf** has the highest basal kinase activity. Activating somatic mutations in *BRAF* are common in melanoma, as well as in colorectal, ovarian, and papillary thyroid carcinomas, underscoring its role as a key oncogenic driver.

The most prevalent *BRAF* mutation in melanoma is **V600E**, which results in constitutive kinase activity and persistent MAPK pathway signaling, promoting uncontrolled cell proliferation.

BRAF mutations are significantly more frequent in melanomas arising on skin with **intermittent sun exposure**. Notably, **81% of melanomas on skin without chronic sun damage** harbor either *BRAF* or *NRAS* mutations, highlighting the importance of MAPK pathway activation in this molecular subtype.